

3(a)	$\rho = m / V$	C1
	$V = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \text{ (= 0.458 m}^3\text{)}$	C1
	$m = \pi \times (0.16 / 2)^2 \times 7.6 \times 3.0 \times 1.2 = 0.55 \text{ kg}$	A1
3(b)(i)	$\Delta p = 0.55 \times 7.6$ $= 4.2 \text{ N s}$	A1
3(b)(ii)	$F = 4.2 / 3.0 \text{ or } 0.55 \times 7.6 / 3.0$ $= 1.4 \text{ N}$	A1
3(c)(i)	$F = 1.4 \text{ N}$	A1
3(c)(ii)	Newton's third law (of motion)	B1
3(d)	$2 \times 1.4 = m \times 9.81$ $m = 0.29 \text{ kg}$	A1
3(e)	the density of air is less at high altitude	B1
3(f)	$f_o = f_s v / (v - v_s)$ $= 3000 \times 340 / (340 - 22)$	C1
	$= 3200 \text{ Hz}$	A1